MICROECONOMICS

MICRO-67

Time: 5 minutes

Outcome: Explain the basic limitation on combining individual rankings into a social ranking.

Difficulty: Intermediate

Arrow's Impossibility Theorem

THE CORE IDEA

With at least three alternatives, Arrow's theorem shows that no ranked-choice social decision rule can guarantee all of a specified set of desirable conditions at once. The result identifies a tradeoff among aggregation properties; it does not prove that democracy or every voting system is impossible.

HOW TO RECOGNIZE IT

- The problem combines individual rankings into a complete and consistent social ranking.
- Desirable conditions include respecting unanimity and avoiding control by one dictator.
- Independence of irrelevant alternatives limits how an outside option may change a pairwise ranking.
- The theorem concerns simultaneous guarantees under its conditions, not whether voting can occur.

WATCH OUT

Do not conclude that no voting system works or that democracy is impossible. The theorem applies to ranked aggregation and the simultaneous satisfaction of specified conditions.

WORKED EXAMPLE: DESIRABLE RULES IN TENSION

DIAGNOSE

- social ranking
- unanimity
- nondictatorship
- consistency
- irrelevant alternatives

A community wants a ranked-choice rule that respects unanimous rankings, is not controlled by one voter, produces a consistent social ranking, and treats irrelevant alternatives appropriately. Each property seems reasonable. Arrow's theorem says that, with at least three alternatives and unrestricted individual rankings, no aggregation rule can guarantee all of them simultaneously. The lesson is an unavoidable design tradeoff, not a claim that collective choice is meaningless.

CHECK YOURSELF

Which is accurate: Arrow identifies incompatible guarantees under stated conditions, or proves that no election can produce a winner?

READY? **Return to the game and master this concept.**